



Covering Dimension of Finite Distributive Lattices

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Abstract

In this paper, we study the covering dimension $\dim_{\mathcal{L}}$ of finite distributive lattices. By the Birkhoff's representation theorem for finite distributive lattices, we prove that for any finite distributive lattice L , $\dim_{\mathcal{L}}(L) = \text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(L)))$, where $\text{Max}(\mathcal{J}(L))$ is the set of all maximal elements of join-irreducible elements of L . Based on the relationships between the covering dimension of topological spaces and that of bounded distributive lattices, we characterize the covering dimension of finite T_0 spaces. Finally, we study the covering dimension of the linear sum, Cartesian product, lexicographic product and rectangular product of two finite distributive lattices.

Keywords Covering dimension · Finite distributive lattice · Linear sum · Cartesian product · Lexicographic product · Rectangular product

1 Introduction

Dimension theory of posets is a branch of order theory that focuses on defining and studying the concept of dimension in certain classes of posets. It plays an important role in the study of posets. There are dimensions of posets, lattices and frames: order dimension, Krull dimension, small inductive dimension, covering dimension, and quasi-covering dimension have been studied in [4, 7–10, 12–14, 18–20, 24, 25]. In [10], Charalambous introduced the notion of σ -frames and established a meaningful dimension theory for normal σ -frames. In [12], Dube et al. developed the theory of covering dimension $\dim_{\mathcal{L}}$ for finite lattices. Especially, they used the notion of minimal covers to characterize the covering dimension for finite lattices. In [25], Zhang et al. proved that $\dim_{\mathcal{L}}(L_1 \times L_2) = \max\{\dim_{\mathcal{L}}(L_1), \dim_{\mathcal{L}}(L_2)\}$ for all finite lattices L_1 and L_2 . They gave a negative answer to the question being that whether the relation $\dim_{\mathcal{L}}(L_1 \circ L_2) \leq \dim_{\mathcal{L}}(L_1) + \dim_{\mathcal{L}}(L_2) + 1$ holds for all finite lattices L_1 and L_2 .

Notably, Dube et al. proved that for any finite lattice L , $\dim_{\mathcal{L}}(L) = \max\{\text{ord}_{\mathcal{L}}(C) \mid C \text{ is a minimal cover of } L\}$. Since we do not have an equivalent characterization of minimal cover in a finite lattice, it is not straightforward to identify all minimal covers of a finite lattice.

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This may bring a challenge in finding a simple method to calculate the covering dimension of finite lattices. Nevertheless, we can deal with this challenge in the setting of finite distributive lattices. This is our main motivation.

In this paper, we focus on the covering dimension of finite distributive lattices. By the Birkhoff's representation theorem for finite distributive lattices, we prove that for every finite distributive lattice L , $\text{Max}(\mathcal{J}(L))$ is the unique minimal cover of L , and then we obtain that $\dim_{\mathcal{L}}(L) = \text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(L)))$. Based on the relationships between the covering dimension of topological spaces and that of bounded distributive lattices, we characterize the covering dimension of finite T_0 spaces. Finally, we give the sum and product theorems of the covering dimension of finite distributive lattices.

2 Covering Dimension of Finite Distributive Lattices

Our references for elementary properties of lattices are [11, 21]. Let P be a poset and $A \subseteq P$. A is called a down-set if, whenever $x \in A$, $y \in P$ and $y \leq x$, we have $y \in A$. For $A \subseteq P$, we write $A^{\downarrow} = \{y \in P \mid y \leq x \text{ for every } x \in A\}$ and $\downarrow x = \{y \in P \mid y \leq x\}$. Dually, we can define up-set, $A^{\uparrow}$ and $\uparrow x$. The set of down-sets of P will be denoted by $\text{Down}(P)$. Obviously, $(\text{Down}(P), \subseteq)$ is a frame. For every $\mathcal{A} \subseteq \text{Down}(P)$, $\bigvee \mathcal{A} = \bigcup \mathcal{A}$, $\bigwedge \mathcal{A} = \bigcap \mathcal{A}$. $\emptyset$ is the bottom element and P is the top element. We use $\text{Down}(P)$ to denote the down-set lattice $(\text{Down}(P), \subseteq)$. It is easy to prove that $\text{Down}(P)$ is a topology of P , we use $D(P)$ to denote the topological space $(P, \text{Down}(P))$.

Let P be a poset and $Q \subseteq P$. If for every $a \in P$, there exists a subset A of Q such that $a = \bigvee A$, then Q is called join-dense in P . We use $\text{Max}(P)$ to denote the set of all the maximal elements of P and $\text{Min}(P)$ to denote the set of all the minimal elements of P .

Let L be a bounded lattice. The top element and the bottom element of L will be denoted by 1_L and 0_L , respectively. L is called prime if $x \wedge y = 0_L$ implies that $x = 0_L$ or $y = 0_L$ for all $x, y \in L$. A subset C of L is said to be a cover of L if $0_L \notin C$ and $\bigvee C = 1_L$. A cover D of L is a refinement of a cover C , written $D \ll C$, if for every $d \in D$, there exists $c \in C$ such that $d \leq c$. It is easy to see that the relation $\ll$ is a quasiorder (reflexive and transitive).

Definition 2.1 (see [10]) Let L be a bounded lattice. The order of a subset C of L is defined as follows:

- (1) $\text{ord}_{\mathcal{L}}(C) = -1$ if and only if $C = \{0_L\}$.
- (2) $\text{ord}_{\mathcal{L}}(C) = k$, where $k \in \mathbb{N}_0$, if the meet of any $k + 2$ distinct elements of C is 0_L and there exist $k + 1$ distinct elements of C whose meet is not 0_L .
- (3) $\text{ord}_{\mathcal{L}}(C) = \infty$ if for every $k \in \mathbb{N}_0$, there exist k distinct elements of C whose meet is not 0_L .

Definition 2.2 (see [10]) Let $\mathbf{BLat}$ be the class of bounded lattices. We define the covering dimension function

$$\dim_{\mathcal{L}}: \mathbf{BLat} \rightarrow \{-1, \infty\} \cup \mathbb{N}_0$$

as follows:

- (1) $\dim_{\mathcal{L}}(L) = -1$ if and only if $L = \{0_L\}$.
- (2) $\dim_{\mathcal{L}}(L) \leq k$, where $k \in \mathbb{N}_0$, if every finite cover C of L has a finite refinement R with $\text{ord}_{\mathcal{L}}(R) \leq k$.
- (3) $\dim_{\mathcal{L}}(L) = k$, where $k \in \mathbb{N}_0$, if $\dim_{\mathcal{L}}(L) \leq k$ and $\dim_{\mathcal{L}}(L) \not\leq k - 1$.
- (4) $\dim_{\mathcal{L}}(L) = \infty$ if $\dim_{\mathcal{L}}(L) \not\leq k$ for any $k \in \{-1\} \cup \mathbb{N}_0$.

Definition 2.3 (see [12]) A cover C of a finite lattice L is minimal if $C \subseteq R$ for every refinement R of C .

Definition 2.4 (see [11]) Let L be a lattice. An element $x \in L$ is join-irreducible if

- (1) $x \neq 0_L$ (in case L has the bottom element 0_L);
- (2) $x = a \vee b$ implies that $x = a$ or $x = b$ for all $a, b \in L$.

We denote the set of all join-irreducible elements of L by $\mathcal{J}(L)$.

Definition 2.5 (see [11]) Let P be a poset and $x, y \in P$. We say x is covered by y (or x is a lower cover of y), and write $x \prec y$, if $x < y$ and $x \leq z < y$ implies $z = x$.

Definition 2.6 (see [11]) Let L be a lattice with the bottom element 0_L . $a \in L$ is called an atom if $0 \prec a$. The set of all atoms of L is denoted by $\text{Atom}(L)$.

Proposition 2.7 (see [11]) Let L be a finite lattice. An element is join-irreducible element if and only if it has exactly one lower cover.

Corollary 2.8 Let L be a finite lattice. For every $a \in \text{Atom}(L)$, a is a join-irreducible element of L .

Proposition 2.9 (see [11]) Let L be a finite lattice. Then $\mathcal{J}(L)$ is join-dense in L .

Proposition 2.10 Let L be a finite lattice and C a minimal cover of L . Then every element of C is a join-irreducible element and any two elements of C are incomparable.

Proof Let C be a minimal cover of L . If $c \in C$ is not a join-irreducible element, by Proposition 2.9, there exists $A \subseteq \mathcal{J}(L)$ such that $\bigvee A = c$, then $(C \setminus \{c\}) \cup A$ is a refinement of C . However, $C \not\subseteq (C \setminus \{c\}) \cup A$, this contradicts with the fact that C is a minimal cover. Suppose that $a, b \in C$ such that $a \leq b$, then $C \setminus \{a\}$ is a refinement of C . However, $C \not\subseteq C \setminus \{a\}$, this contradicts with the fact that C is a minimal cover. $\square$

Proposition 2.11 Let L be a finite lattice. Then for every $A \subseteq L$, $\bigwedge A = 0_L$ if and only if $A^l \cap \mathcal{J}(L) = \emptyset$.

Proof If $\bigwedge A = 0_L$, then $A^l \cap \mathcal{J}(L) = \emptyset$. So the necessity holds. Conversely, suppose that $A^l \cap \mathcal{J}(L) = \emptyset$ and there exists $x \in L$ such that $x \neq 0_L$ and $x \leq a$ for every $a \in A$. By Proposition 2.9, there exists a non-empty subset $B \subseteq \mathcal{J}(L)$ such that $\bigvee B = x$. Then $B \subseteq A^l$, but this contradicts with the fact that $A^l \cap \mathcal{J}(L) = \emptyset$. So we do not have $x \neq 0_L$ as a common lower bound of A , the sufficiency holds. $\square$

Proposition 2.12 (see [12]) Let L be a finite lattice. Then

$$\dim_{\mathcal{L}}(L) = \max\{\text{ord}_{\mathcal{L}}(C) \mid C \text{ is a minimal cover of } L\}.$$

Example 2.13 Obviously, the diamond lattice $\mathbf{M}_3 = \{\perp, \top, a, b, c\}$ is a finite lattice, where a, b and c are incomparable elements, and $\perp, \top$ are the bottom element and the top element, respectively. We can see that $\{a, b\}$, $\{b, c\}$ and $\{a, c\}$ are the minimal covers of $\mathbf{M}_3$. Then

$$\dim_{\mathcal{L}}(L) = \max\{\text{ord}_{\mathcal{L}}(\{a, b\}), \text{ord}_{\mathcal{L}}(\{b, c\}), \text{ord}_{\mathcal{L}}(\{a, c\})\} = 0.$$

By Example 2.13, we can see that the minimal cover of a finite lattice is not unique. Next, we will prove that the minimal cover of a finite distributive lattice is unique.

Proposition 2.14 (see [6]) (Birkhoff's representation theorem for finite distributive lattices) *Let L be a finite distributive lattice. Then the map*

$$\varphi : a \mapsto \text{spec}(a)$$

is an isomorphism between L and $\text{Down}(\mathcal{J}(L))$, where $\text{spec}(a) = \{x \in \mathcal{J}(L) \mid x \leq a\} = \downarrow a \cap \mathcal{J}(L)$ is the spectrum of a .

Proposition 2.15 *Let L be a finite distributive lattice. Then $\text{Max}(\mathcal{J}(L))$ is the unique minimal cover of L .*

Proof It is easy to see that $\{\text{spec}(a) \mid a \in \text{Max}(\mathcal{J}(L))\}$ is a cover of $\text{Down}(\mathcal{J}(L))$. Let $\mathcal{C}$ be a cover of $\text{Down}(\mathcal{J}(L))$. As $\bigvee \mathcal{C} = \bigcup \mathcal{C} = \mathcal{J}(L)$, for each $a \in \text{Max}(\mathcal{J}(L))$, there exists $C \in \mathcal{C}$ such that $a \in \text{spec}(a) \subseteq C$. Thus, $\{\text{spec}(a) \mid a \in \text{Max}(\mathcal{J}(L))\}$ is a refinement of every cover of $\text{Down}(\mathcal{J}(L))$. Let cover $\mathcal{D}$ be a refinement of $\{\text{spec}(a) \mid a \in \text{Max}(\mathcal{J}(L))\}$. By the definition of refinement, for every $D \in \mathcal{D}$, there exists $a \in \text{Max}(\mathcal{J}(L))$ such that $D \subseteq \text{spec}(a)$. However, if there exist $D' \in \mathcal{D}$ and $a' \in \text{Max}(\mathcal{J}(L))$ such that $D' \subsetneq \text{spec}(a')$, $a' \notin D'$, and $a' \notin \bigcup \mathcal{D}$, this contradicts with the fact that $\mathcal{D}$ is a cover of $\text{Down}(\mathcal{J}(L))$. So the refinement of $\{\text{spec}(a) \mid a \in \text{Max}(\mathcal{J}(L))\}$ is itself. Therefore, we have that $\{\text{spec}(a) \mid a \in \text{Max}(\mathcal{J}(L))\}$ is a minimal cover of $\text{Down}(\mathcal{J}(L))$. Let $\mathcal{C}$ be a minimal cover of $\text{Down}(\mathcal{J}(L))$. As $\{\text{spec}(a) \mid a \in \text{Max}(\mathcal{J}(L))\}$ is a refinement of $\mathcal{C}$, $\mathcal{C} \subseteq \{\text{spec}(a) \mid a \in \text{Max}(\mathcal{J}(L))\}$. But $\{\text{spec}(a) \mid a \in \text{Max}(\mathcal{J}(L))\}$ is a minimal cover of $\text{Down}(\mathcal{J}(L))$. So $\mathcal{C} = \{\text{spec}(a) \mid a \in \text{Max}(\mathcal{J}(L))\}$. Therefore, we can conclude that $\{\text{spec}(a) \mid a \in \text{Max}(\mathcal{J}(L))\}$ is the unique minimal cover of $\text{Down}(\mathcal{J}(L))$.

By Proposition 2.14, $\varphi : a \mapsto \text{spec}(a)$ is an isomorphism between L to $\text{Down}(\mathcal{J}(L))$, then

$$\varphi^{-1}(\{\text{spec}(a) \mid a \in \text{Max}(\mathcal{J}(L))\}) = \bigvee \{\text{spec}(a) \mid a \in \text{Max}(\mathcal{J}(L))\} = \text{Max}(\mathcal{J}(L))$$

is the unique minimal cover of L . $\square$

By Propositions 2.12 and 2.15, we can obtain the following theorem.

Theorem 2.16 *Let L be a finite distributive lattice. Then $\dim_{\mathcal{L}}(L) = \text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(L)))$.*

Remark 2.17 By Proposition 2.15, when L is a finite distributive lattice, the minimal cover of L is unique. However, let us consider the pentagon lattice $\mathbf{N}_5 = \{\perp, \top, u, v, w\}$, where $u \leq v$, u and v are incomparable with w , and $\perp, \top$ are the bottom element and the top element, respectively. Then $\mathbf{N}_5$ is not a distributive lattice. It is obvious that the minimal cover of $\mathbf{N}_5$ is unique. So distributiveness is not a necessary condition for the uniqueness of the minimal cover.

By Proposition 2.11 and Theorem 2.16, we can obtain the following proposition.

Proposition 2.18 *Let L be a finite distributive lattice. Then $\dim_{\mathcal{L}}(L) = 0$ if and only if $\text{spec}(a) \cap \text{spec}(b) = \emptyset$ for every $a, b \in \text{Max}(\mathcal{J}(L))$.*

Proposition 2.19 *Let L be a finite distributive lattice and $\dim_{\mathcal{L}}(L) = k$ with $k \geq 1$. Then $|L| \geq 2^{k+1} + 1$.*

Proof Suppose that L is a finite distributive lattice and $\dim_{\mathcal{L}}(L) = k$ with $k \geq 1$. By Theorem 2.16, $\mathcal{J}(L)$ must satisfy:

(i) the meet of any $k + 2$ distinct elements of $\text{Max}(\mathcal{J}(L))$ is 0_L ,

(ii) there exist $k + 1$ distinct elements of $\text{Max}(\mathcal{J}(L))$ whose meet is not 0_L .

By Proposition 2.11, (i) and (ii) are equivalent to (i') and (ii'), respectively.

(i') there is no join-irreducible element serving as a lower bound for any $k + 2$ distinct elements of $\text{Max}(\mathcal{J}(L))$,

(ii') there exist $k + 1$ distinct elements of $\text{Max}(\mathcal{J}(L))$ that share a common join-irreducible element as their lower bound.

Then $|\mathcal{J}(L)| \geq k + 2$ and $\mathcal{J}(L)$ must have a subposet $P = \{x_0, x_1, \dots, x_{k+1}\}$, where the Hasse diagram of P is Fig. 1. By Proposition 2.14, we can conclude that $|L| \geq |\text{Down}(P)|$. As $\text{Down}(P) = \{\emptyset\} \cup \{\{x_0\} \cup A \mid A \subseteq \{x_1, x_2, \dots, x_{k+1}\}\}$, we have that $|\text{Down}(P)| = 2^{k+1} + 1$. So $|L| \geq 2^{k+1} + 1$. $\square$

Corollary 2.20 *Let L be a finite distributive lattice with n elements. Then $\dim_{\mathcal{L}}(L) \leq \lceil \log_2(n - 1) - 1 \rceil$, where $\lceil \cdot \rceil$ is the ceiling function.*

Example 2.21 Let L be a finite distributive lattice. The Hasse diagram of L is Fig. 2.

Then the Hasse diagram of $\mathcal{J}(L)$ is Fig. 3.

It is easy to see that $\text{Max}(\mathcal{J}(L)) = \{m, o\}$. Then $\dim_{\mathcal{L}}(L) = \text{ord}_{\mathcal{L}}(\{m, o\}) = 1$.

3 Covering Dimension of Finite T_0 Spaces

The covering dimension of finite T_0 spaces has been studied extensively (see [2, 3, 15, 16, 22]). In this section, we will investigate the covering dimension of finite T_0 spaces based on some properties regarding the covering dimension of finite lattices.

Let X be a topological space. A cover of X is a non-empty set of subsets of X , whose union is X . A cover $\mathcal{A}$ of X is said to be open if all elements of $\mathcal{A}$ are open. A cover $\mathcal{A}$ of X is said to be a refinement of a cover $\mathcal{C}$ of X if for every $A \in \mathcal{A}$ there exists $C \in \mathcal{C}$ such that $A \subseteq C$.

Let X be an Alexandroff space. For every $x \in X$, we use U_x to denote the smallest open set containing x .

Definition 3.1 (see [1]) Let X be a T_0 space. The specialization order $\leq$ on X is defined as follows:

$$x \leq y \text{ iff } x \in cl(\{y\}),$$

where $cl(\{y\})$ is the closure of $\{y\}$.

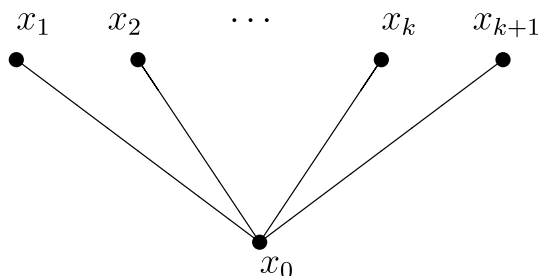


Fig. 1 Hasse diagram of P

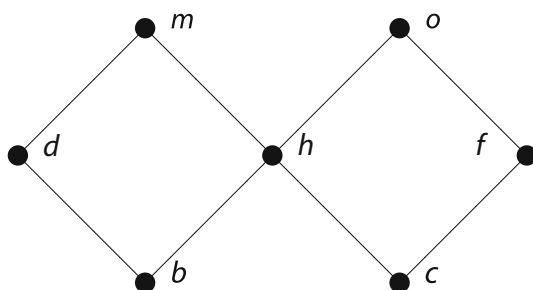


Fig. 3 Hasse diagram of $\mathcal{J}(L)$

$\dim_{\mathcal{T}} : \mathbf{F} \longrightarrow \{-1, \infty\} \cup \mathbb{N}_0$
as follows:

- (1) $\dim_{\mathcal{T}}(X) \leq k$, where $k \in \{-1\} \cup \mathbb{N}_0$, iff for every finite open cover $\mathcal{C}$ of the space X , there exists a finite open cover $\mathcal{A}$ of X , refinement of $\mathcal{C}$, such that $\text{ord}_{\mathcal{T}}(\mathcal{A}) \leq k$.
- (2) $\dim_{\mathcal{T}}(X) = k$, where $k \in \mathbb{N}_0$, iff $\dim_{\mathcal{T}}(X) \leq k$ and $\dim_{\mathcal{T}}(X) \not\leq k - 1$.
- (3) $\dim_{\mathcal{T}}(X) = \infty$ iff there exists no $k \in \{-1\} \cup \mathbb{N}_0$ such that $\dim_{\mathcal{T}}(X) \leq k$.

Definition 3.5 (see [11]) Let L be a lattice. A subset F of L is called an filter if

- (1) $a, b \in F$ implies $a \wedge b \in F$;
- (2) $a \in L, b \in F$ and $b \leq a$ imply $a \in F$.

Definition 3.6 (see [11]) Let L be a lattice. A filter F of L is said to be prime if for every $a, b \in L$, $a \vee b \in F$ implies $a \in F$ or $b \in F$.

The set of all prime filters of L is denoted by $\mathbf{P}(L)$. We write $h(a) = \{F \in \mathbf{P}(L) \mid a \in F\}$. Then h is an isomorphism of L with $h(L)$, whose join and meet are set union and intersection, respectively. $\mathbf{P}(L)$ is given the topology which has $h(L)$ as a base.

Proposition 3.7 (see [10]) Let L be a bounded distributive lattice. Then $\dim_{\mathcal{L}}(L) = \dim_{\mathcal{T}}(\mathbf{P}(L))$.

It is well known that there is a one-to-one correspondence between join-irreducible elements and prime filters when L is a finite distributive lattice. By Proposition 3.7, we have the following proposition.

Proposition 3.8 Let L be a finite distributive lattice. Then $\dim_{\mathcal{L}}(L) = \dim_{\mathcal{T}}(\mathbf{D}(\mathcal{J}(L)))$.

Proof It suffices to prove that the topological spaces $\mathbf{P}(L)$ and $\mathbf{D}(\mathcal{J}(L))$ are homeomorphic when L is a finite distributive lattice. It can be easily seen that the map $\phi : P \longmapsto \bigwedge P$ is a bijection between $\mathbf{P}(L)$ and $\mathcal{J}(L)$. Then $\phi^{-1}(p) = \uparrow p$ for every $p \in \mathcal{J}(L)$.

Claim 1. ϕ is continuous from $\mathbf{P}(L)$ to $\mathbf{D}(\mathcal{J}(L))$.

Let U be an open set of $\mathbf{D}(\mathcal{J}(L))$. By Proposition 2.14, there exists $x \in L$ such that $U = \text{spec}(x)$. We can conclude that $\phi^{-1}(\text{spec}(x)) = h(x)$. So $\phi^{-1}(U)$ is an open set of $\mathbf{P}(L)$.

Claim 2. ϕ^{-1} is continuous from $\mathbf{D}(\mathcal{J}(L))$ to $\mathbf{P}(L)$.

Let $\mathcal{U}$ be an open set of $\mathbf{P}(L)$. Since $h(L)$ is a base of $\mathbf{P}(L)$, there exists a subset $A \subseteq L$ such that $\mathcal{U} = \bigcup_{a \in A} h(a)$. Since ϕ is a bijection, we have that

$$\phi(\mathcal{U}) = \phi\left(\bigcup_{a \in A} h(a)\right) = \bigcup_{a \in A} \phi(h(a)) = \bigcup_{a \in A} \text{spec}(a) = \text{spec}\left(\bigvee A\right).$$

Then $\phi(\mathcal{U})$ is an open set of $D(\mathcal{J}(L))$.

Therefore, ϕ is a homeomorphism between $P(L)$ and $D(\mathcal{J}(L))$. $\square$

Proposition 3.9 (see [10]) *Let X be a topological space. Then $\dim_{\mathcal{T}}(X) = \dim_{\mathcal{L}}(\mathcal{O}(X))$, where $\mathcal{O}(X)$ is the open set lattice of X .*

Proposition 3.10 *Let X be a finite T_0 space. Then the following statements hold:*

- (1) $\mathcal{J}(\mathcal{O}(X)) = \{U_x \mid x \in X\}$;
- (2) $\text{Max}(\mathcal{J}(\mathcal{O}(X))) = \{U_x \mid x \in \text{Min}(X)\}$.

Proof (1) For every $x \in X$ and $U, V \in \mathcal{O}(X)$, if $U_x = U \cup V$, then $x \in U_x \subseteq U$ or $x \in U_x \subseteq V$. Since $U \subseteq U_x$ and $V \subseteq U_x$, we have that $U_x = U$ or $U_x = V$. Thus we have that U_x is a join-irreducible element of $\mathcal{O}(X)$. Conversely, let U be a join-irreducible element of $\mathcal{O}(X)$. As $U = \bigcup_{x \in U} U_x$ and U is a finite set, there exists $x \in U$ such that $U = U_x$. So $\mathcal{J}(\mathcal{O}(X)) = \{U_x \mid x \in X\}$.

- (2) By Remark 3.2, $U_x \in \text{Max}(\mathcal{J}(\mathcal{O}(X))) \iff U_x \not\subseteq U_y$ for each $y \neq x$
 $\iff y \not\subseteq x$ for each $y \neq x$
 $\iff x \in \text{Min}(X)$.

So we can conclude that $\text{Max}(\mathcal{J}(\mathcal{O}(X))) = \{U_x \mid x \in \text{Min}(X)\}$. $\square$

Theorem 3.11 *Let X be a finite T_0 space. Then $\dim_{\mathcal{T}}(X) = \text{ord}_{\mathcal{T}}(\{U_x \mid x \in \text{Min}(X)\})$.*

Proof $\dim_{\mathcal{T}}(X) = \dim_{\mathcal{L}}(\mathcal{O}(X))$
 $= \text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(\mathcal{O}(X))))$ (By Theorem 2.16)
 $= \text{ord}_{\mathcal{L}}(\{U_x \mid x \in \text{Min}(X)\})$ (By Proposition 3.10)
 $= \text{ord}_{\mathcal{T}}(\{U_x \mid x \in \text{Min}(X)\})$. $\square$

Remark 3.12 By Theorem 3.11, we can also conclude that the algorithm in [23] for computing the covering dimension of finite T_0 spaces is feasible.

4 Sum and Products Theorems

Definition 4.1 (see [11]) Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be disjoint posets. The disjoint union $L_1 \uplus L_2$ is the poset $(L_1 \cup L_2, \leq)$, where the relation $\leq$ is defined as follows:

$$x \leq y \iff \begin{cases} x, y \in L_1 \text{ and } x \leq_1 y \text{ or} \\ x, y \in L_2 \text{ and } x \leq_2 y. \end{cases}$$

Definition 4.2 (see [11]) Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be disjoint posets. The linear sum $L_1 \oplus L_2$ is the poset $(L_1 \cup L_2, \leq)$, where the relation $\leq$ is defined as follows:

$$x \leq y \iff \begin{cases} x, y \in L_1 \text{ and } x \leq_1 y \text{ or} \\ x, y \in L_2 \text{ and } x \leq_2 y \text{ or} \\ x \in L_1, y \in L_2. \end{cases}$$

It is easy to see that if $(L_1, \leq_1)$ and $(L_2, \leq_2)$ are lattices, then $L_1 \oplus L_2$ is a lattice.

Proposition 4.3 *Let L_1 and L_2 be distributive lattices. Then $L_1 \oplus L_2$ is a distributive lattice.*

Proof It suffices to prove that $L_1 \oplus L_2$ does not have a sublattice isomorphic to $\mathbf{M}_3$ or $\mathbf{N}_5$. Let $L_1 \oplus L_2$ have a sublattice M isomorphic to $\mathbf{M}_3$, where $M = \{p, q, u, v, w\}$ and p, q are the bottom element and the top element, respectively. By Definition 4.2, we have that $\{u, v, w\} \subseteq L_1$ or $\{u, v, w\} \subseteq L_2$. Then $M \subseteq L_1$ or $M \subseteq L_2$, M is a sublattice of L_1 or M is a sublattice of L_2 , which contradicts with the fact that L_1 and L_2 are distributive lattices. Let $L_1 \oplus L_2$ have a sublattice N isomorphic to $\mathbf{N}_5$, where $N = \{s, t, x, y, z\}$ and s, t are the top element and bottom element, respectively. By Definition 4.2, we have that $\{x, y, z\} \subseteq L_1$ or $\{x, y, z\} \subseteq L_2$. Then $N \subseteq L_1$ or $N \subseteq L_2$, N is a sublattice of L_1 or N is a sublattice of L_2 , which contradicts with the fact that L_1 and L_2 are distributive lattices. So $L_1 \oplus L_2$ is a distributive lattice. $\square$

Proposition 4.4 Let L be a finite lattice. If $L = \{0_L\} \oplus (L_1 \uplus L_2) \oplus \{1_L\}$ (see Fig. 4), then

$$\dim_{\mathcal{L}}(L) = \max\{\dim_{\mathcal{L}}(\{0_L\} \oplus L_1 \oplus \{1_L\}), \dim_{\mathcal{L}}(\{0_L\} \oplus L_2 \oplus \{1_L\})\}.$$

Proof Let C be a minimal cover of L . If $C \subseteq \{0_L\} \oplus L_1 \oplus \{1_L\}$, then C is also a minimal cover of $\{0_L\} \oplus L_1 \oplus \{1_L\}$, $\text{ord}_{\mathcal{L}}(C) \leq \dim_{\mathcal{L}}(\{0_L\} \oplus L_1 \oplus \{1_L\})$. Similarly, if $C \subseteq \{0_L\} \oplus L_2 \oplus \{1_L\}$, we have that $\text{ord}_{\mathcal{L}}(C) \leq \dim_{\mathcal{L}}(\{0_L\} \oplus L_2 \oplus \{1_L\})$. If $C \not\subseteq \{0_L\} \oplus L_1 \oplus \{1_L\}$, then $C = \{a, b\}$, where $a \in L_1$ and $b \in L_2$, $\text{ord}_{\mathcal{L}}(\{a, b\}) = 0$. Otherwise, without loss of generality, if $C \supseteq \{a, b, c\}$ such that $\{a, c\} \subseteq L_1$ and $b \in L_2$, by Definition 4.1, we have that $a \vee b = 1_L$. Then $\{a, b\}$ is a refinement of C , which contradicts with the fact that C is a minimal cover of L . Therefore, $\dim_{\mathcal{L}}(L) = \max\{\text{ord}_{\mathcal{L}}(C) \mid C \text{ is a minimal cover of } L\} \leq \max\{\dim_{\mathcal{L}}(\{0_L\} \oplus L_1 \oplus \{1_L\}), \dim_{\mathcal{L}}(\{0_L\} \oplus L_2 \oplus \{1_L\})\}$.

Let C be a minimal cover of $\{0_L\} \oplus L_1 \oplus \{1_L\}$ and R be a refinement of C in L . If $C \neq \{1_L\}$, by the definition of refinement, we have that $R \subseteq L_1$. Then R is a refinement of C in $\{0_L\} \oplus L_1 \oplus \{1_L\}$. As C is a minimal cover of $\{0_L\} \oplus L_1 \oplus \{1_L\}$, $C \subseteq R$. Then C is a minimal cover of L . If $C = \{1_L\}$, then $\text{ord}_{\mathcal{L}}(C) = 0$. Thus $\dim_{\mathcal{L}}(\{0_L\} \oplus L_1 \oplus \{1_L\}) \leq \dim_{\mathcal{L}}(L)$. Similarly, we can prove that $\dim_{\mathcal{L}}(\{0_L\} \oplus L_2 \oplus \{1_L\}) \leq \dim_{\mathcal{L}}(L)$. Then $\max\{\dim_{\mathcal{L}}(\{0_L\} \oplus$

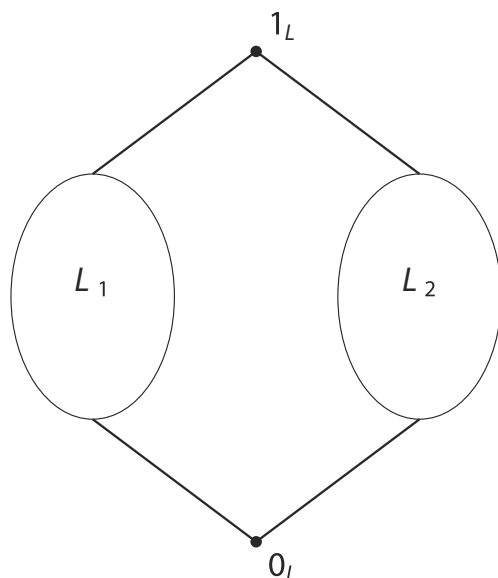


Fig. 4 $L = \{0_L\} \oplus (L_1 \uplus L_2) \oplus \{1_L\}$

$L_1 \oplus \{1_L\}$, $\dim_{\mathcal{L}}(\{0_L\} \oplus L_2 \oplus \{1_L\}) \leq \max\{\text{ord}_{\mathcal{L}}(C) \mid C \text{ is a minimal cover of } L\} = \dim_{\mathcal{L}}(L)$.

Therefore, $\dim_{\mathcal{L}}(L) = \max\{\dim_{\mathcal{L}}(\{0_L\} \oplus L_1 \oplus \{1_L\}), \dim_{\mathcal{L}}(\{0_L\} \oplus L_2 \oplus \{1_L\})\}$. $\square$

Proposition 4.5 *Let L_1 and L_2 be bounded lattices. Then the following statements hold:*

- (1) $0_{L_2} \in \mathcal{J}(L_1 \oplus L_2)$;
- (2) If $a \in \mathcal{J}(L_1)$, then $a \in \mathcal{J}(L_1 \oplus L_2)$;
- (3) If $b \in \mathcal{J}(L_2)$, then $b \in \mathcal{J}(L_1 \oplus L_2)$;
- (4) $\mathcal{J}(L_1 \oplus L_2) = \{0_{L_2}\} \cup \mathcal{J}(L_1) \cup \mathcal{J}(L_2)$.

Proof (1) Assume that there exist $x, y \in L_1 \cup L_2$ such that $0_{L_2} = x \vee y$, but $x \neq 0_{L_2}$ and $y \neq 0_{L_2}$. By Definition 4.2, we have that $x, y \in L_1$. Therefore, $x \vee y \leq 1_{L_1} < 0_{L_2}$, which contradicts with the fact that $0_{L_2} = x \vee y$.

(2) Let $x, y \in L_1 \cup L_2$ such that $a = x \vee y$. If $x \in L_2$ or $y \in L_2$, then $a < 0_{L_2} \leq x \vee y$, which contradicts with the fact that $a = x \vee y$. Therefore, $x, y \in L_1$. Since $a \in \mathcal{J}(L_1)$, we have that $x = a$ or $y = a$. So $a \in \mathcal{J}(L_1 \oplus L_2)$.

(3) Let $x, y \in L_1 \cup L_2$ such that $b = x \vee y$. If x and y are both in L_1 , then $x \vee y \leq 1_{L_1} < b$, which contradicts with the fact that $b = x \vee y$. Therefore, $x \in L_2$ or $y \in L_2$. If $x, y \in L_2$, since $b \in \mathcal{J}(L_2)$, we have that $x = b$ or $y = b$. Then $b \in \mathcal{J}(L_1 \oplus L_2)$. If $x \in L_1$ and $y \in L_2$, then $b = x \vee y = y$, and thus $b \in \mathcal{J}(L_1 \oplus L_2)$. If $x \in L_2$ and $y \in L_1$, then $b = x \vee y = x$. So $b \in \mathcal{J}(L_1 \oplus L_2)$.

(4) By (1), (2) and (3), we have that $\{0_{L_2}\} \cup \mathcal{J}(L_1) \cup \mathcal{J}(L_2) \subseteq \mathcal{J}(L_1 \oplus L_2)$. It suffices to prove that $\mathcal{J}(L_1 \oplus L_2) \subseteq \{0_{L_2}\} \cup \mathcal{J}(L_1) \cup \mathcal{J}(L_2)$. Let $c \in \mathcal{J}(L_1 \oplus L_2)$ and $c \neq 0_{L_2}$. If $c \in L_1$, then for every $x, y \in L_1$, when $c = x \vee y$, we have that $c = x$ or $c = y$. Thus $c \in \mathcal{J}(L_1)$. If $c \in L_2$, then for every $x, y \in L_2$, when $c = x \vee y$, we have that $c = x$ or $c = y$. Thus $c \in \mathcal{J}(L_2)$. Therefore, $\mathcal{J}(L_1 \oplus L_2) \subseteq \{0_{L_2}\} \cup \mathcal{J}(L_1) \cup \mathcal{J}(L_2)$. $\square$

Theorem 4.6 *Let L_1 and L_2 be finite distributive lattices. Then*

$$\dim_{\mathcal{L}}(L_1 \oplus L_2) = |\text{Max}(\mathcal{J}(L_2))| - 1.$$

Proof By Definition 4.2 and Proposition 4.5, we have that $\text{Max}(\mathcal{J}(L_1 \oplus L_2)) = \text{Max}(\mathcal{J}(L_2))$. Since 0_{L_2} is a common lower bound of $\text{Max}(\mathcal{J}(L_2))$, according to Theorem 2.16, we have that $\dim_{\mathcal{L}}(L_1 \oplus L_2) = \text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(L_2))) = |\text{Max}(\mathcal{J}(L_2))| - 1$. $\square$

Definition 4.7 (see [11]) Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be posets. The Cartesian product $L_1 \times L_2$ is the poset $(L_1 \times L_2, \leq)$, where the relation $\leq$ is defined as follows:

$$(x_1, x_2) \leq (y_1, y_2) \iff x_1 \leq_1 y_1, x_2 \leq_2 y_2.$$

It is easy to see that if L_1 and L_2 are lattices, then $L_1 \times L_2$ is a lattice, where $(x_1, x_2) \vee (y_1, y_2) = (x_1 \vee y_1, x_2 \vee y_2)$ and $(x_1, x_2) \wedge (y_1, y_2) = (x_1 \wedge y_1, x_2 \wedge y_2)$ for every $(x_1, x_2), (y_1, y_2) \in L_1 \times L_2$.

Proposition 4.8 *Let L_1 and L_2 be distributive lattices. Then $L_1 \times L_2$ is a distributive lattice.*

Proposition 4.9 *Let L_1 and L_2 be bounded lattices. Then*

$$\mathcal{J}(L_1 \times L_2) = \{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\} \cup \{(0_{L_1}, y) \mid y \in \mathcal{J}(L_2)\}.$$

Proof Let $x \in \mathcal{J}(L_1)$ and suppose that there exist $(a_1, b_1), (a_2, b_2) \in L_1 \times L_2$ such that $(a_1, b_1) \vee (a_2, b_2) = (x, 0_{L_2})$. As $(a_1, b_1) \vee (a_2, b_2) = (a_1 \vee a_2, b_1 \vee b_2) = (x, 0_{L_2})$,

$a_1 \vee a_2 = x$ and $b_1 \vee b_2 = 0_{L_2}$. Thus we have that $a_1 = x$ or $a_2 = x$, $b_1 = b_2 = 0_{L_2}$. So $(x, 0_{L_2})$ is a join-irreducible element of $L_1 \times L_2$.

It can also be proved that $(0_{L_1}, y)$ is a join-irreducible element of $L_1 \times L_2$ when $y \in \mathcal{J}(L_2)$.

If $x = 0_{L_1}$ and $y \notin \mathcal{J}(L_2)$, then there exist $a, b \in L_2$ such that $a, b \neq y$ and $a \vee b = y$. Since $(0_{L_1}, a) \neq (0_{L_1}, y)$ and $(0_{L_1}, b) \neq (0_{L_1}, y)$, but $(0_{L_1}, a) \vee (0_{L_1}, b) = (0_{L_1}, y)$, we can conclude that $(0_{L_1}, y)$ is not a join-irreducible element of $L_1 \times L_2$.

It can also be proved that $(x, 0_{L_2})$ is not a join-irreducible element of $L_1 \times L_2$ when $x \notin \mathcal{J}(L_1)$.

If $x \neq 0_{L_1}$ and $y \neq 0_{L_2}$, then $(x, 0_{L_2}) \vee (0_{L_1}, y) = (x, y)$. So (x, y) is not a join-irreducible element of $L_1 \times L_2$. $\square$

Theorem 4.10 *Let L_1 and L_2 be finite distributive lattices. Then*

$$\dim_{\mathcal{L}}(L_1 \times L_2) = \max\{\text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(L_1))), \text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(L_2)))\}.$$

Proof By Theorem 2.16 and Proposition 4.9, we suffice to prove that

$$\begin{aligned} & \text{ord}_{\mathcal{L}}(\text{Max}(\{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\} \cup \{(0_{L_1}, y) \mid y \in \mathcal{J}(L_2)\})) \\ &= \max\{\text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(L_1))), \text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(L_2)))\}. \end{aligned}$$

Suppose that $\text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(L_1))) = k_1$, $\text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(L_2))) = k_2$ and $k_1 \geq k_2$. Since $\text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(L_1))) = k_1$, there exist $k_1 + 1$ distinct elements $\{x_1, x_2, \dots, x_{k_1+1}\} \subseteq \text{Max}(\{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\})$ such that $\bigwedge \{x_1, x_2, \dots, x_{k_1+1}\} \neq (0_{L_1}, 0_{L_2})$. By Definition 4.7, we have that a and b is incomparable for every $a \in \{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\}$ and every $b \in \{(0_{L_1}, y) \mid y \in \mathcal{J}(L_2)\}$. Then

$$\begin{aligned} & \text{Max}(\{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\} \cup \{(0_{L_1}, y) \mid y \in \mathcal{J}(L_2)\}) \\ &= \text{Max}(\{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\}) \cup \text{Max}(\{(0_{L_1}, y) \mid y \in \mathcal{J}(L_2)\}). \end{aligned}$$

Therefore,

$$\{x_1, x_2, \dots, x_{k_1+1}\} \subseteq \text{Max}(\{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\} \cup \{(0_{L_1}, y) \mid y \in \mathcal{J}(L_2)\}).$$

So we find the $k_1 + 1$ distinct elements of $\text{Max}(\{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\} \cup \{(0_{L_1}, y) \mid y \in \mathcal{J}(L_2)\})$ such that its meet is not $(0_{L_1}, 0_{L_2})$.

For any $k_1 + 2$ distinct elements $\{x_1, x_2, \dots, x_{k_1+2}\} \subseteq \text{Max}(\{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\} \cup \{(0_{L_1}, y) \mid y \in \mathcal{J}(L_2)\})$, if $\{x_1, x_2, \dots, x_{k_1+2}\} \subseteq \text{Max}(\{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\})$ or $\{x_1, x_2, \dots, x_{k_1+2}\} \subseteq \text{Max}(\{(0_{L_1}, y) \mid y \in \mathcal{J}(L_2)\})$, by the definition of $\text{ord}_{\mathcal{L}}$, we have that $\bigwedge \{x_1, x_2, \dots, x_{k_1+2}\} = (0_{L_1}, 0_{L_2})$. If there exist $x_i \in \text{Max}(\{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\})$ and $x_j \in \text{Max}(\{(0_{L_1}, y) \mid y \in \mathcal{J}(L_2)\})$, then $\bigwedge \{x_1, x_2, \dots, x_{k_1+2}\} \leq x_i \wedge x_j = (0_{L_1}, 0_{L_2})$. So the meet of any $k_1 + 2$ distinct elements $\{x_1, x_2, \dots, x_{k_1+2}\} \subseteq \text{Max}(\{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\} \cup \{(0_{L_1}, y) \mid y \in \mathcal{J}(L_2)\})$ is $(0_{L_1}, 0_{L_2})$. $\square$

Definition 4.11 (see [11]) Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be lattices. The lexicographic product $L_1 \circ L_2$ is the poset $(L_1 \times L_2, \leq)$ where the relation $\leq$ is defined as follows:

$$(x_1, x_2) \leq (y_1, y_2) \iff \begin{cases} x_1 <_1 y_1 \text{ or} \\ x_1 = y_1 \text{ and } x_2 \leq_2 y_2. \end{cases}$$

Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be bounded lattices. It is easy to see that

$$(x_1, x_2) \wedge (y_1, y_2) = \begin{cases} (x_1, x_2), & \text{if } x_1 <_1 y_1 \\ (y_1, y_2), & \text{if } y_1 <_1 x_1 \\ (x_1, x_2 \wedge y_2), & \text{if } x_1 = y_1 \\ (x_1 \wedge y_1, 1_{L_2}), & \text{if } x_1 \parallel y_1 \end{cases}$$

and

$$(x_1, x_2) \vee (y_1, y_2) = \begin{cases} (x_1, x_2), & \text{if } y_1 <_1 x_1 \\ (y_1, y_2), & \text{if } x_1 <_1 y_1 \\ (x_1, x_2 \vee y_2), & \text{if } x_1 = y_1 \\ (x_1 \vee y_1, 0_{L_2}), & \text{if } x_1 \parallel y_1. \end{cases}$$

Remark 4.12 Let L_1 be a Boolean lattice with four elements, and let L_2 be a chain with two elements. It is obvious that L_1 and L_2 are distributive lattices, but $L_1 \circ L_2$ is not a distributive lattice. So the lexicographic product of bounded distributive lattices is not necessarily distributive lattice.

Proposition 4.13 *Let L_1 and L_2 be bounded lattices. If $x \in \mathcal{J}(L_1)$, then $(x, 0_{L_2}) \in \mathcal{J}(L_1 \circ L_2)$.*

Proof Suppose that there exist $(a_1, a_2), (b_1, b_2) \in L_1 \circ L_2$ such that $(x, 0_{L_2}) = (a_1, a_2) \vee (b_1, b_2)$.

- (i) If $b_1 < a_1$, then $(x, 0_{L_2}) = (a_1, a_2) \vee (b_1, b_2) = (a_1, a_2)$.
- (ii) If $a_1 < b_1$, then $(x, 0_{L_2}) = (a_1, a_2) \vee (b_1, b_2) = (b_1, b_2)$.
- (iii) If $a_1 = b_1$, then $(x, 0_{L_2}) = (a_1, a_2) \vee (b_1, b_2) = (a_1, a_2 \vee b_2)$, $(x, 0_{L_2}) = (a_1, a_2)$.
- (iv) If $a_1 \parallel b_1$, then $(x, 0_{L_2}) = (a_1, a_2) \vee (b_1, b_2) = (a_1 \vee b_1, 0_{L_2})$. Since $x \in \mathcal{J}(L_1)$, $x = a_1$ or $x = b_1$. However, this contradicts with the fact that $a_1 \parallel b_1$, so this situation does not exist.

Then we can conclude that $(x, 0_{L_2})$ is a join-irreducible element of $L_1 \circ L_2$. □

Proposition 4.14 *Let L_1 and L_2 be bounded lattices. Then $\{(x, y) \mid x \in L_1, y \in \mathcal{J}(L_2)\} \subseteq \mathcal{J}(L_1 \circ L_2)$.*

Proof Suppose that $x \in L_1, y \in \mathcal{J}(L_2)$ and there exist $(a_1, a_2), (b_1, b_2) \in L_1 \circ L_2$ such that $(a_1, a_2) \vee (b_1, b_2) = (x, y)$.

- (i) If $b_1 < a_1$, then $(a_1, a_2) \vee (b_1, b_2) = (a_1, a_2) = (x, y)$.
- (ii) If $a_1 < b_1$, then $(a_1, a_2) \vee (b_1, b_2) = (b_1, b_2) = (x, y)$.
- (iii) If $a_1 = b_1$, then $(a_1, a_2) \vee (b_1, b_2) = (a_1, a_2 \vee b_2) = (x, y)$. As $y \in \mathcal{J}(L_2)$, $(a_1, a_2) = (x, y)$ or $(b_1, b_2) = (x, y)$.
- (iv) If $a_1 \parallel b_1$, then $(x, y) = (a_1, a_2) \vee (b_1, b_2) = (a_1 \vee b_1, 0_{L_2})$. Since $0_{L_2} \notin \mathcal{J}(L_2)$, this situation does not exist.

So we can conclude that (x, y) is a join-irreducible element of $L_1 \circ L_2$. □

Proposition 4.15 *Let L_1 and L_2 be bounded lattices. Then*

$$\mathcal{J}(L_1 \circ L_2) = \{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\} \cup \{(x, y) \mid x \in L_1, y \in \mathcal{J}(L_2)\}.$$

Proof By Propositions 4.13 and 4.14, we have that

$$\{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\} \cup \{(x, y) \mid x \in L_1, y \in \mathcal{J}(L_2)\} \subseteq \mathcal{J}(L_1 \circ L_2).$$

If $x \notin \mathcal{J}(L_1)$, then there exist $a, b \in L_1$ such that $x = a \vee b$ but $x \neq a, x \neq b$. Thus, we have that $(x, 0_{L_2}) = (a, 0_{L_2}) \vee (b, 0_{L_2})$. Since $(x, 0_{L_2}) \neq (a, 0_{L_2})$ and $(x, 0_{L_2}) \neq (b, 0_{L_2})$, $(x, 0_{L_2}) \notin \mathcal{J}(L_1 \circ L_2)$ when $x \notin \mathcal{J}(L_1)$.

If $x \in L_1, y \neq 0_{L_2}$ and $y \notin \mathcal{J}(L_2)$, then there exist $a, b \in L_2$ such that $y = a \vee b$ but $y \neq a$ and $y \neq b$. Thus, we have that $(x, y) = (x, a) \vee (x, b)$. Since $(x, a) \neq (x, y)$ and $(x, b) \neq (x, y)$, $(x, y) \notin \mathcal{J}(L_1 \circ L_2)$ when $y \neq 0_{L_2}$ and $y \notin \mathcal{J}(L_2)$.

Therefore, we prove that $a \notin \mathcal{J}(L_1 \circ L_2)$ when $a \notin \{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\}$ and $a \notin \{(x, y) \mid x \in L_1, y \in \mathcal{J}(L_2)\}$. So $\mathcal{J}(L_1 \circ L_2) = \{(x, 0_{L_2}) \mid x \in \mathcal{J}(L_1)\} \cup \{(x, y) \mid x \in L_1, y \in \mathcal{J}(L_2)\}$. $\square$

Theorem 4.16 Let L_1 be a finite lattice and let L_2 be a finite distributive lattice. Then

$$\dim_{\mathcal{L}}(L_1 \circ L_2) = |\text{Max}(\mathcal{J}(L_2))| - 1.$$

Proof Since L_2 is a finite distributive lattice, $\{1_{L_1}\} \circ L_2$ is a finite distributive lattice. As

$$\text{Max}(\mathcal{J}(\{1_{L_1}\} \circ L_2)) = \{(1_{L_1}, y) \mid y \in \text{Max}(\mathcal{J}(L_2))\},$$

by Proposition 2.15, $\{(1_{L_1}, y) \mid y \in \text{Max}(\mathcal{J}(L_2))\}$ is the unique minimal cover of $\{1_{L_1}\} \circ L_2$. It is obvious that $\{(1_{L_1}, y) \mid y \in \text{Max}(\mathcal{J}(L_2))\}$ is a cover of $L_1 \circ L_2$. Let C be a cover of $L_1 \circ L_2$ such that $C \ll \{(1_{L_1}, y) \mid y \in \text{Max}(\mathcal{J}(L_2))\}$. By Definition 4.11, we have that $(1_{L_1}, 0_{L_2})$ is an upper bound of $C \cap ((L_1 \circ L_2) \setminus (\{1_{L_1}\} \circ L_2))$. Then

$$\bigvee C \cap (\{1_{L_1}\} \circ L_2) = (1_{L_1}, 1_{L_2}),$$

and thus $C \cap (\{1_{L_1}\} \circ L_2)$ is a cover of $\{1_{L_1}\} \circ L_2$. Since

$$C \cap (\{1_{L_1}\} \circ L_2) \ll C \ll \{(1_{L_1}, y) \mid y \in \text{Max}(\mathcal{J}(L_2))\},$$

we have that

$$\{(1_{L_1}, y) \mid y \in \text{Max}(\mathcal{J}(L_2))\} \subseteq C \cap (\{1_{L_1}\} \circ L_2) \subseteq C.$$

So $\{(1_{L_1}, y) \mid y \in \text{Max}(\mathcal{J}(L_2))\}$ is a minimal cover of $L_1 \circ L_2$.

Let D be a minimal cover of $L_1 \circ L_2$. Assume that $D \not\subseteq \{1_{L_1}\} \circ L_2$. Then

$$D \cap ((L_1 \circ L_2) \setminus (\{1_{L_1}\} \circ L_2)) \neq \emptyset.$$

By Definition 4.11, we have that $(1_{L_1}, 0_{L_2})$ is an upper bound of $D \cap ((L_1 \circ L_2) \setminus (\{1_{L_1}\} \circ L_2))$. Then

$$\bigvee D \cap (\{1_{L_1}\} \circ L_2) = (1_{L_1}, 1_{L_2}).$$

Clearly, $D \cap (\{1_{L_1}\} \circ L_2) \ll D$. Since D is a minimal cover, we can conclude that $D \subseteq D \cap (\{1_{L_1}\} \circ L_2)$. This contradicts with the fact that $D \cap ((L_1 \circ L_2) \setminus (\{1_{L_1}\} \circ L_2)) \neq \emptyset$. So $D \subseteq \{1_{L_1}\} \circ L_2$. Let F be a refinement of D in $\{1_{L_1}\} \circ L_2$. Then F is a cover of $L_1 \circ L_2$. Since D is a minimal cover of $L_1 \circ L_2$, we have that $D \subseteq F$. So D is also a minimal cover of $\{1_{L_1}\} \circ L_2$. As $\{(1_{L_1}, y) \mid y \in \text{Max}(\mathcal{J}(L_2))\}$ is the unique minimal cover of $\{1_{L_1}\} \circ L_2$, we can obtain that

$$D = \{(1_{L_1}, y) \mid y \in \text{Max}(\mathcal{J}(L_2))\}.$$

Thus $\{(1_{L_1}, y) \mid y \in \text{Max}(\mathcal{J}(L_2))\}$ is the unique minimal cover of $L_1 \circ L_2$.

As $(1_{L_1}, 0_{L_2})$ is a lower bound of $\{(1_{L_1}, y) \mid y \in \text{Max}(\mathcal{J}(L_2))\}$,

$$\dim_{\mathcal{L}}(L_1 \circ L_2) = \text{ord}_{\mathcal{L}}(\{(1_{L_1}, y) \mid y \in \text{Max}(\mathcal{J}(L_2))\}) = |\text{Max}(\mathcal{J}(L_2))| - 1.$$

□

Definition 4.17 (see [5]) Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be bounded lattices. The rectangular product $L_1 \square L_2$ is the poset $(L_1 \square L_2, \leq)$, where

$$L_1 \square L_2 = \{(x, y) \in L_1 \times L_2 \mid x \neq 0_{L_1} \text{ and } y \neq 0_{L_2}\} \cup \{(0_{L_1}, 0_{L_2})\}$$

and the relation $\leq$ is defined as follows:

$$(x_1, x_2) \leq (y_1, y_2) \iff x_1 \leq_1 y_1 \text{ and } x_2 \leq_2 y_2,$$

for every $(x_1, x_2), (y_1, y_2) \in (L_1 \square L_2, \leq)$.

It is easy to see that

$$(x_1, x_2) \wedge (y_1, y_2) = \begin{cases} (x_1 \wedge y_1, x_2 \wedge y_2), & \text{if } x_1 \wedge y_1 \neq 0_{L_1} \text{ and } x_2 \wedge y_2 \neq 0_{L_2} \\ (0_{L_1}, 0_{L_2}), & \text{otherwise} \end{cases}$$

and $(x_1, x_2) \vee (y_1, y_2) = (x_1 \vee y_1, x_2 \vee y_2)$.

Proposition 4.18 Let L_1 and L_2 be finite lattices. Then

$$\mathcal{J}(L_1 \square L_2) = \{(x, b) \mid x \in \mathcal{J}(L_1), b \in \text{Atom}(L_2)\} \cup \{(a, y) \mid a \in \text{Atom}(L_1), y \in \mathcal{J}(L_2)\}.$$

Proof Let $x \in \mathcal{J}(L_1)$ and $b \in \text{Atom}(L_2)$. If there exist $(a_1, b_1), (a_2, b_2) \in L_1 \square L_2$ such that $(a_1, b_1) \vee (a_2, b_2) = (x, b)$, then by Definition 4.17, we have that $(a_1, b_1) \vee (a_2, b_2) = (a_1 \vee a_2, b_1 \vee b_2) = (x, b)$, $a_1 \vee a_2 = x$ and $b_1 \vee b_2 = b$. Since $x \in \mathcal{J}(L_1)$, we have that $a_1 = x$ or $a_2 = x$, $b_1 = b_2 = b$. So (x, b) is a join-irreducible element of $L_1 \square L_2$.

It can also be proved that (a, y) is a join-irreducible element of $L_1 \square L_2$ when $a \in \text{Atom}(L_1)$ and $y \in \mathcal{J}(L_2)$.

If $a \notin \text{Atom}(L_1)$ and $b \notin \text{Atom}(L_2)$, then there exist $a' \in \text{Atom}(L_1)$ such that $a' < a$ and $b' \in \text{Atom}(L_2)$ such that $b' < b$. Therefore, we have that $(a, b') \vee (a', b) = (a, b)$, (a, b) is not a join-irreducible element of $L_1 \square L_2$.

If $a \in \text{Atom}(L_1)$ and $y \notin \mathcal{J}(L_2)$, then there exist $c, d \in L_2$ such that $c, d \neq y$ and $c \vee d = y$. Since $(a, c) \neq (a, y)$, $(a, d) \neq (a, y)$ and $(a, c) \vee (a, d) = (a, y)$, we have that (a, y) is not a join-irreducible element of $L_1 \square L_2$.

It can also be proved that (x, b) is not a join-irreducible element of $L_1 \square L_2$ when $x \notin \mathcal{J}(L_1)$ and $b \in \text{Atom}(L_2)$.

It is obvious that $(0_{L_1}, 0_{L_2})$ is not a join-irreducible element of $L_1 \square L_2$. □

Proposition 4.19 (see [5]) Let L_1 and L_2 be bounded lattices. Then the following statements are equivalent:

- (1) $L_1 \square L_2$ is distributive.
- (2) L_1 and L_2 are distributive and
 - (a) L_1 or L_2 is isomorphic to **2** (**2** is a chain with two elements) or
 - (b) L_1 and L_2 are prime.

Proposition 4.20 Let L be a finite lattice. Then L is prime if and only if $|\text{Atom}(L)| = 1$.

Proof Suppose that L is prime. As L is a finite lattice, $\text{Atom}(L) \neq \emptyset$. Assume that $|\text{Atom}(L)| > 1$. Then there exist two distinct elements $a, b \in \text{Atom}(L)$ such that $a \wedge b = 0_L$, which contradicts with the fact that L is prime. So $|\text{Atom}(L)| = 1$. Conversely, assume that L is not prime. Then there exist $a \neq 0_L$ and $b \neq 0_L$ such that $a \wedge b = 0_L$. Since L is a finite lattice, there exist $\{a_1, a_2, \dots, a_m \mid m \in \mathbb{N}_0\}$, $\{b_1, b_2, \dots, b_n \mid n \in \mathbb{N}_0\} \subseteq L$ such that $0_L < a_1 < a_2 \cdots < a_m = a$ and $0_L < b_1 < b_2 \cdots < b_n = b$. If $a_1 = b_1$, then $0_L < a_1 \leq a \wedge b$, which contradicts with the fact that $a \wedge b = 0_L$. So $a_1 \neq b_1$. It is obvious that $a_1, b_1 \in \text{Atom}(L)$. However, this contradicts with the fact that $|\text{Atom}(L)| = 1$. Thus L is prime. $\square$

Theorem 4.21 *Let L_1 and L_2 be finite prime distributive lattices. Then*

$$\dim_{\mathcal{L}}(L_1 \square L_2) = |\text{Max}(\mathcal{J}(L_1))| + |\text{Max}(\mathcal{J}(L_2))| - 1.$$

Proof By Proposition 4.20, we can conclude that there is only one atom in each of L_1 and L_2 , which we denote as a_1 and a_2 respectively. Then we have that $\mathcal{J}(L_1 \square L_2) = \{(x, a_2) \mid x \in \mathcal{J}(L_1)\} \cup \{(a_1, y) \mid y \in \mathcal{J}(L_2)\}$. Since atom is a join-irreducible element, (a_1, a_2) is the smallest element of $\{(x, a_2) \mid x \in \mathcal{J}(L_1)\} \cup \{(a_1, y) \mid y \in \mathcal{J}(L_2)\}$. By Definition 4.17, we can conclude that for every $a \in \{(x, a_2) \mid x \in \mathcal{J}(L_1)\}$ and every $b \in \{(a_1, y) \mid y \in \mathcal{J}(L_2)\}$ such that $a, b \neq (a_1, a_2)$, a and b are incomparable. Then

$$\begin{aligned} & \text{Max}(\{(x, a_2) \mid x \in \mathcal{J}(L_1)\} \cup \{(a_1, y) \mid y \in \mathcal{J}(L_2)\}) \\ &= \text{Max}(\{(x, a_2) \mid x \in \mathcal{J}(L_1)\}) \cup \text{Max}(\{(a_1, y) \mid y \in \mathcal{J}(L_2)\}). \end{aligned}$$

As (a_1, a_2) is the common lower bound of $\text{Max}(\{(x, a_2) \mid x \in \mathcal{J}(L_1)\}) \cup \text{Max}(\{(a_1, y) \mid y \in \mathcal{J}(L_2)\})$ and $L_1 \square L_2$ is a finite distributive lattice, by Theorem 2.16, we have that

$$\begin{aligned} & \dim_{\mathcal{L}}(L_1 \square L_2) \\ &= \text{ord}_{\mathcal{L}}(\text{Max}(\mathcal{J}(L_1 \square L_2))) \\ &= \text{ord}_{\mathcal{L}}(\text{Max}(\{(x, a_2) \mid x \in \mathcal{J}(L_1)\}) \cup \text{Max}(\{(a_1, y) \mid y \in \mathcal{J}(L_2)\})) \\ &= |\text{Max}(\mathcal{J}(L_1))| + |\text{Max}(\mathcal{J}(L_2))| - 1. \end{aligned} \quad \square$$

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Declarations

Conflict of Interest None

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